САНКТ-ПЕТЕРБУРГСКИЙ ПОЛИТЕХНИЧЕСКИЙ УНИВЕРСТИТ ПЕТРА ВЕДИКОГО
ИНСТИТУТ МАШИНОСТРОЕНИЕ МАТЕРИАЛОВ И ТРАНСПОРТА
ВЫСШАЯ ШКОЛА АВТОМАТИЗАЦИИ И РОБОТОТЕХНИКИ

(подпись преподавателя)

«__» _______________ 2025 г.

ПОЯСНИИТЕЛЬНАЯ ЗАПИСКА К ПРАКТИЧЕСКОМУ ЗАДАНИЮ

ТАУ 3331506/30101

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Оценка: ……………………………………………………….

Комиссия: ……………………………………………………….

……………………………………………………….

Санкт-Петербург
2025 г.

Семинар 1

1.

2 | $3\ddot{y} + 24\dot{y} + 96y = 1(t)$, при $y(0) = 0, \dot{y}(0) = 0$

1.1.

$3\ddot{y} + 24\dot{y} + 96y = 1(t) \qquad y(0)=0,\ \dot{y}(0)=0$

Laplace transform

$L\{3\}\cdot L\{\ddot{y}(t)\} + L\{24\}\cdot L\{\dot{y}(t)\} + L\{96\}\cdot L\{y(t)\} =$

$= L\{1(t)\}$

$L\{\ddot{y}(t)\} = s^2Y(s) - s\cdot y(0) - \dot{y}(0) = s^2Y(s)$

$L\{\dot{y}(t)\} = s\cdot Y(s) - y(0) = sY(s)$

$L\{y(t)\} = Y(s)$

$L\{1(t)\} = \frac{1}{s}$

$3s^2Y(s) + 24\,sY(s) + 96\,Y(s) = \frac{1}{s}$

$Y(s)[3s^2+24s+96] = \frac{1}{s} \quad |: [3s^2+24s+96]$

$Y(s) = \frac{1}{3s^2+24s+96}\cdot\frac{1}{s} = \frac{1}{3(s^2+8s+32)}\cdot\frac{1}{s} =$

$= \frac{1}{3((s^2+8s+16)+16)}\cdot\frac{1}{s} = \frac{1}{3((s+4)^2+16)}\cdot\frac{1}{s} =$

$= \frac{A}{s} + \frac{Bs+C}{3(s^2+8s+32)} \quad |\cdot 3s(s^2+8s+32)$

$1 = 3A(s^2+8s+32) + (Bs+C)s$

$1 = 3A\,s^2 + 24A\cdot s + A\cdot 96 + Bs^2 + Cs$

$1 = s^2(3A+B) + s(24A+C) + 96A$

$s^2: 3A+B=0$

$s: 24A+C=0$

$s^0: 96A=1$

$\Rightarrow \quad A = \frac{1}{96},\quad B = -\frac{1}{32},\quad C = -\frac{1}{4}$

$$Y(s) = \frac{1}{96}\cdot\frac{1}{s} - \frac{1}{32}\cdot\frac{s}{3(s^2+8s+32)} - \frac{1}{4}\cdot\frac{1}{3(s^2+8s+32)}$$

$$s^2+8s+32=(s^2+8s+16)+16 = (s+4)^2+4^2$$

$$-\frac{1}{96}\left(\frac{s+8}{(s+4)^2+4^2}\right) = -\frac{1}{96}\left(\frac{s+4}{(s+4)^2+4^2}+\frac{4}{(s+4)^2+4^2}\right)$$

$$y(t) = \frac{1}{96}(t) - \frac{1}{96}\cdot e^{-4t}(\cos(4t)+\sin(4t))$$

1.2.

```
-   y1=simplify(dsolve('3*D2y+24*Dy+96*y=1','y(0)=0','Dy(0)=0'))
```

mmand Window

```
y1 =

1/96 - (2^(1/2)*sin(4*t + pi/4)*exp(-4*t))/96

>>
```

1.3.

```
syms s
y2=1/(s*(3*s^2+24*s+96));
y2=ilaplace(y2)
```

Command Window

```
y2 =

1/96 - (exp(-4*t)*(cos(4*t) + sin(4*t)))/96

fx >>
```

1.4.

1,4 $$Y(s)\cdot(3s^2+24s+96) = U(s).$$

$$Y(s) = \underbrace{\frac{U(s)}{3s^2+24s+96}}_{\text{Force response}} + \underbrace{\frac{\cancel{0} \nearrow 0}{3s^2+24s+96}}_{\text{Free response}}$$

$$\frac{Y(s)}{U(s)} = H(s) = \frac{1}{3s^2+24s+96}$$

2.

2	$\ddot{y} + \dot{y} + 6y = 2u$

$$s^2 \cdot Y(s) + s \cdot Y(s) + 6\, Y(s) = 2U(s)$$

1) $u(t) = \delta(t)$

$$s^2 Y(s) + s\, Y(s) + 6\, Y(s) = 2 \cdot 1$$

$$Y(s) = \frac{2}{(s^2 + s + 6)}$$

(2) $u(t) = 1(t)$

$$s^2 Y(s) + s\, Y(s) + 6\, Y(s) = \frac{2}{s}$$

$$Y(s) = \frac{2}{s(s^2 + s + 6)}$$

2.1.

```
syms s
y3=2/(s^2+s+6);
y3=ilaplace(y3)

y4=2/(s*(s^2+s+6));
y4=ilaplace(y4)
```

```
y3 =

(4*23^(1/2)*exp(-t/2)*sin((23^(1/2)*t)/2))/23


y4 =

1/3 - (exp(-t/2)*(cos((23^(1/2)*t)/2) + (23^(1/2)*sin((23^(1/2)*t)/2))/23))/3
```

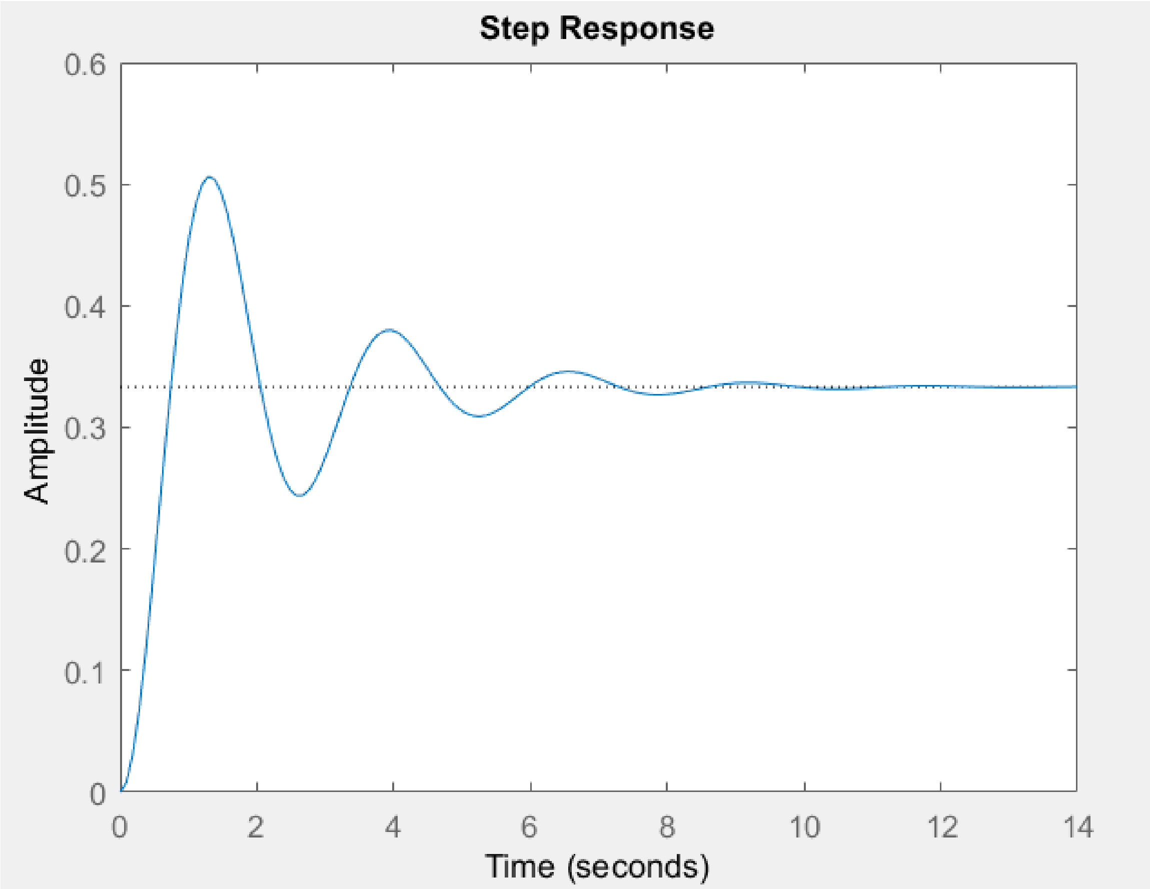
2.2.

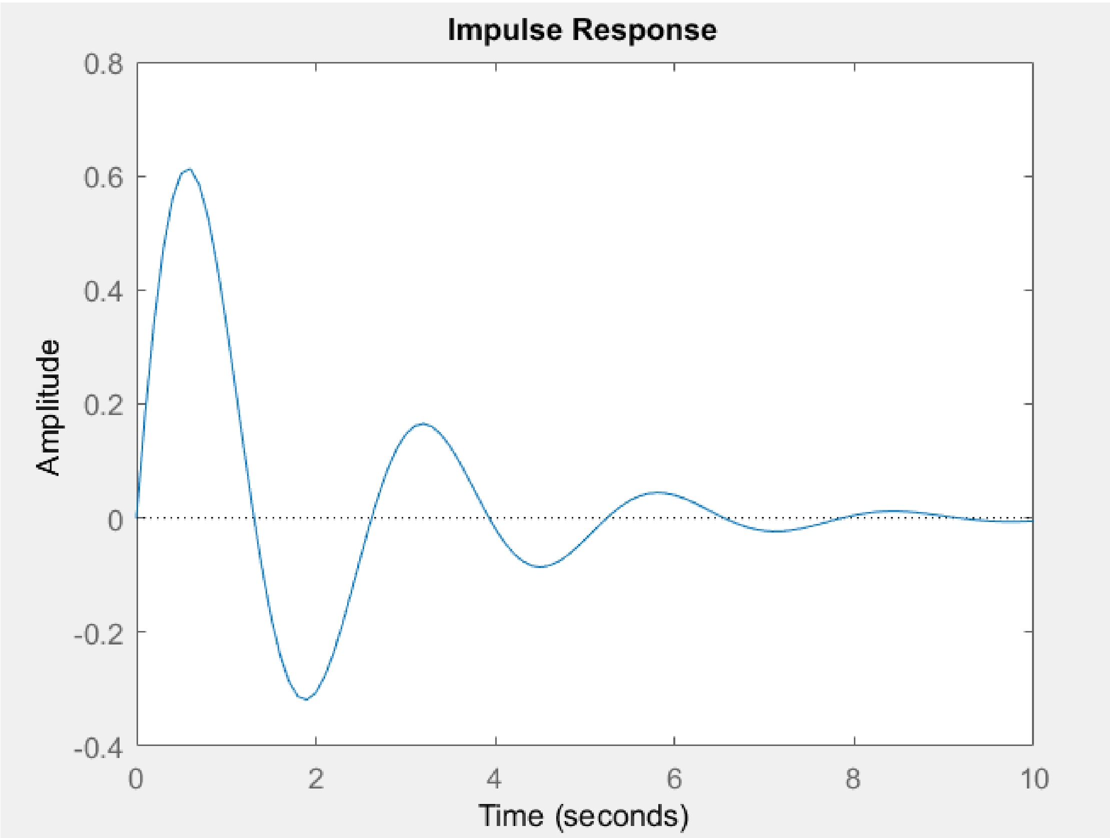
$$\frac{Y(s)}{U(s)} = H(s) = \frac{2}{s^2+s+6}$$

$$h(t) = \frac{4\cdot\sqrt{23}\cdot e^{-\frac{t}{2}}\cdot \sin\left(\frac{\sqrt{23}\,t}{2}\right)}{23}$$

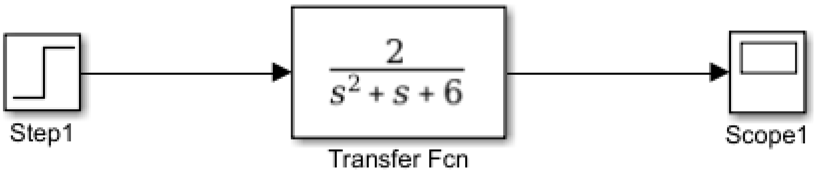
2.3.

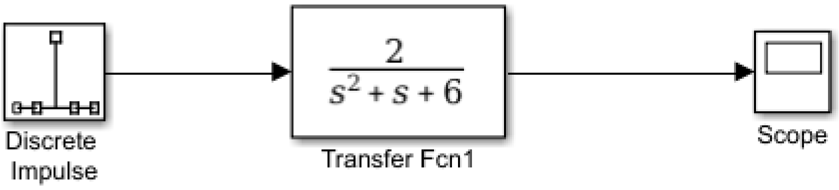
```
num = [2];
den = [1 1 6];
h = tf(num,den);

hold on
step(h)
impulse(h,10)
```

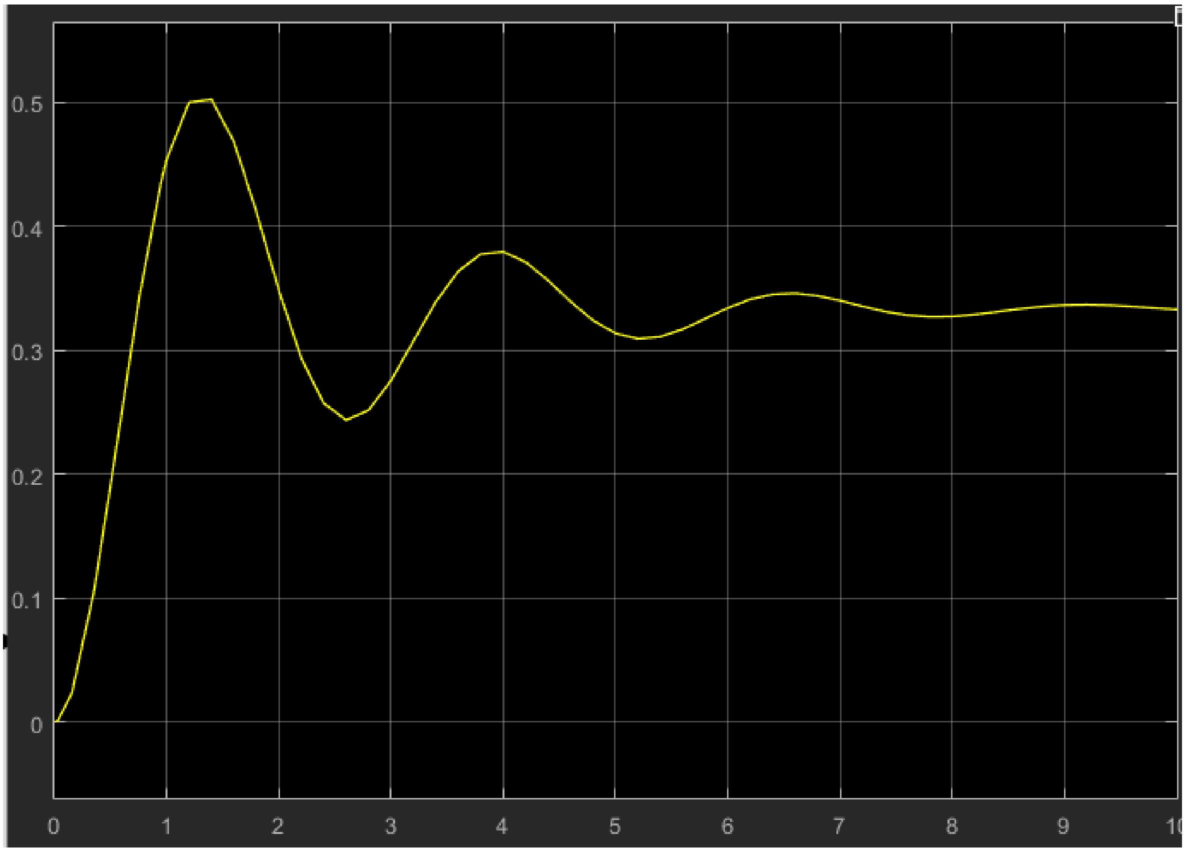

Step Response
Amplitude
Time (seconds)
0.6
0.5
0.4
0.3
0.2
0.1
0
0
2
4
6
8
10
12
14

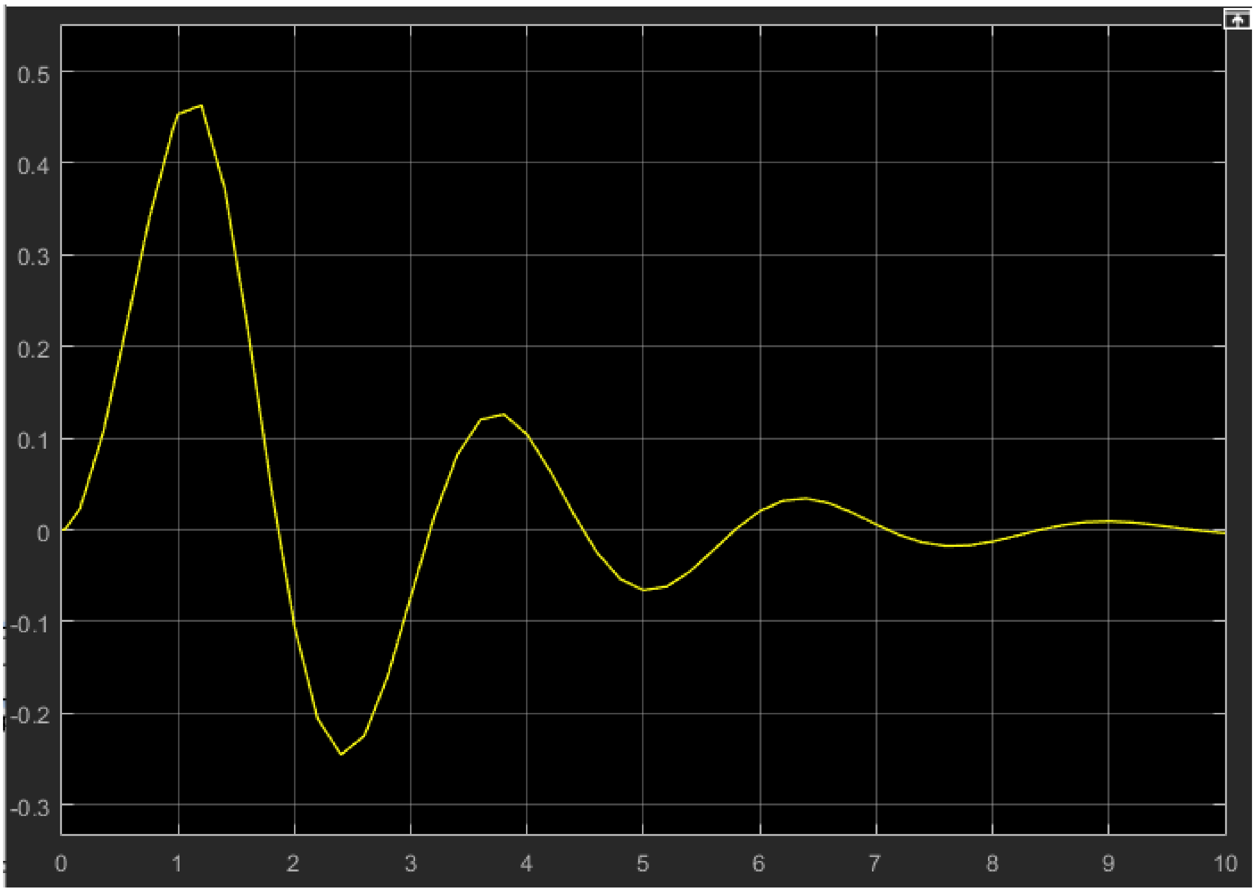

Impulse Response
Amplitude
Time (seconds)
0.8
0.6
0.4
0.2
0
-0.2
-0.4
0
2
4
6
8
10

2.4.


Step1
$\frac{2}{s^2+s+6}$
Transfer Fcn
Scope1


Discrete
Impulse
$\frac{2}{s^2+s+6}$
Transfer Fcn1
Scope


0.5
0.4
0.3
0.2
0.1
0
0
1
2
3
4
5
6
7
8
9

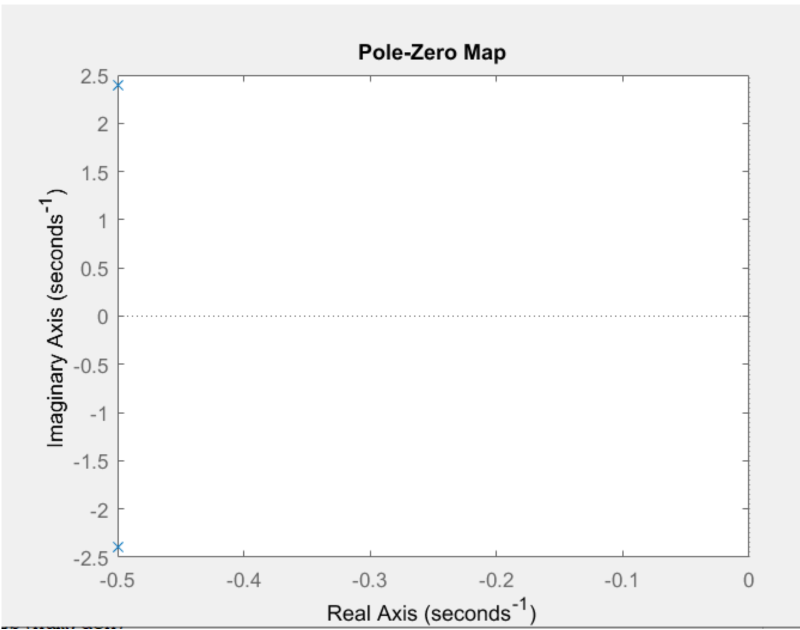


0.5
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
0
1
2
3
4
5
6
7
8
9
10

2.5.

```
S = tf('s');
num = 2;
den=(S^2+S+6);
a=(num/den);

get(a);
num = [2];
den = [1 1 6];
[Z,P,K]=tf2zp(num,den)
pzmap(a)
```

```
Z =

  0×1 empty double column vector

P =

  -0.5000 + 2.3979i
  -0.5000 - 2.3979i

K =

     2
```



Stable

Bounded

y1 = (4*23^(1/2)*exp(-t/2)*sin((23^(1/2)*t)/2))/23

bounded

y2 = 1/3 - (exp(-t/2)*(cos((23^(1/2)*t)/2) + (23^(1/2)*sin((23^(1/2)*t)/2))/23))/3

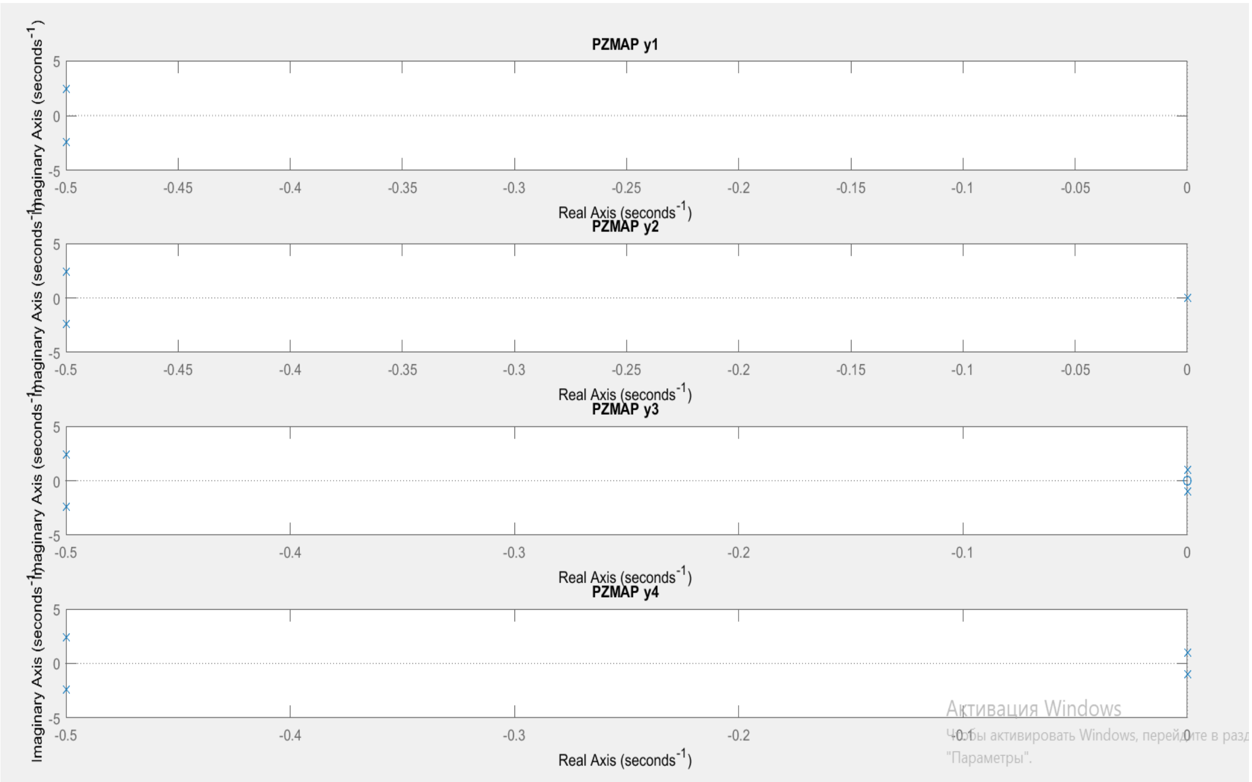
bounded

y3=(5*cos(t))/13 + sin(t)/13 - (5*exp(-t/2)*(cos((23^(1/2)*t)/2) + (7*23^(1/2)*sin((23^(1/2)*t)/2))/115))/13

bounded

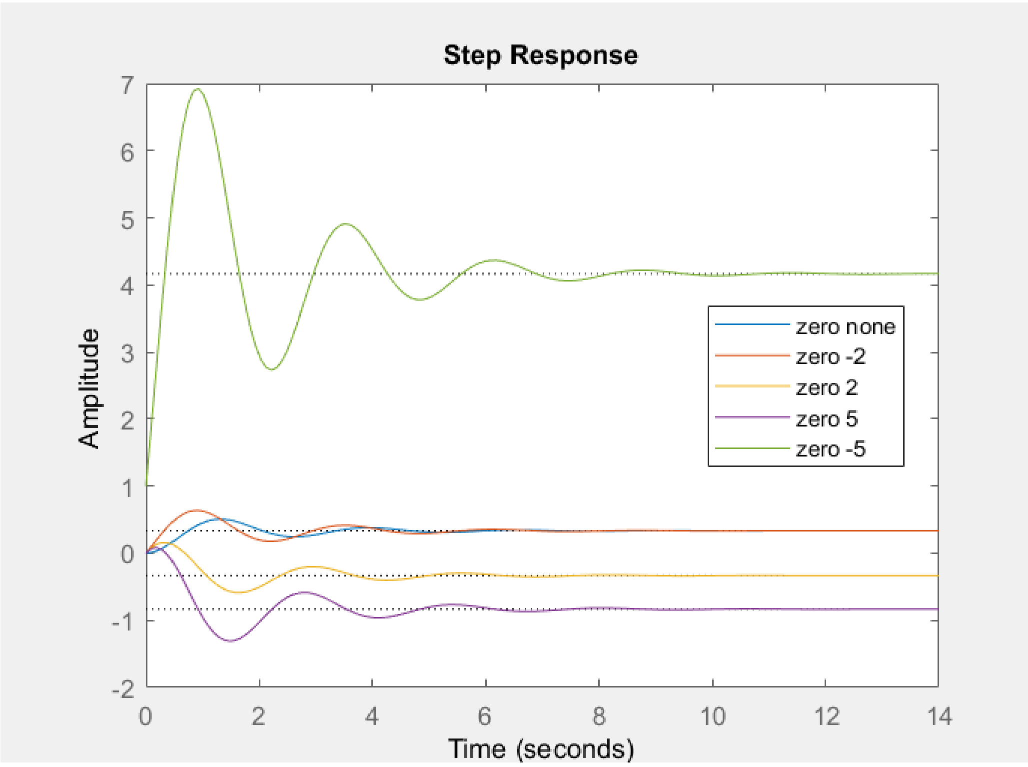
y4 = (5*sin(t))/13 - cos(t)/13 + (exp(-t/2)*(cos((23^(1/2)*t)/2) - (9*23^(1/2)*sin((23^(1/2)*t)/2))/23))/13

bounded



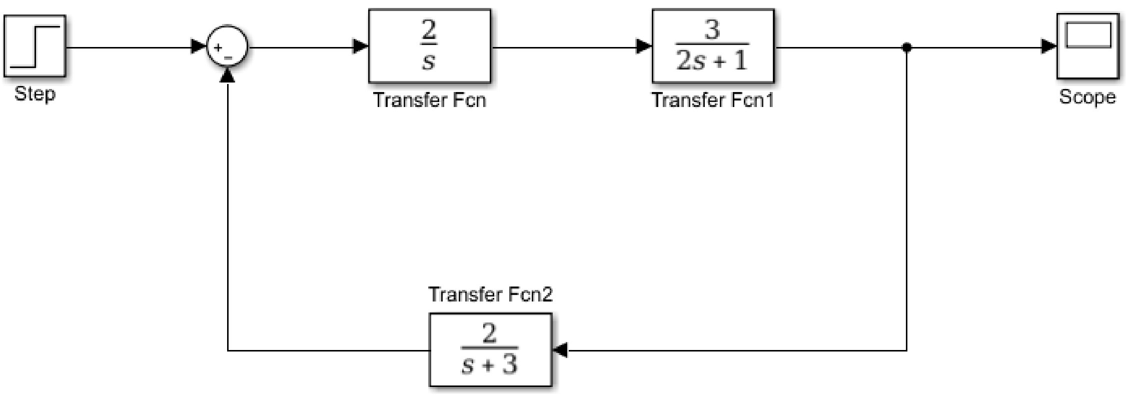
clc; clear all; close all;

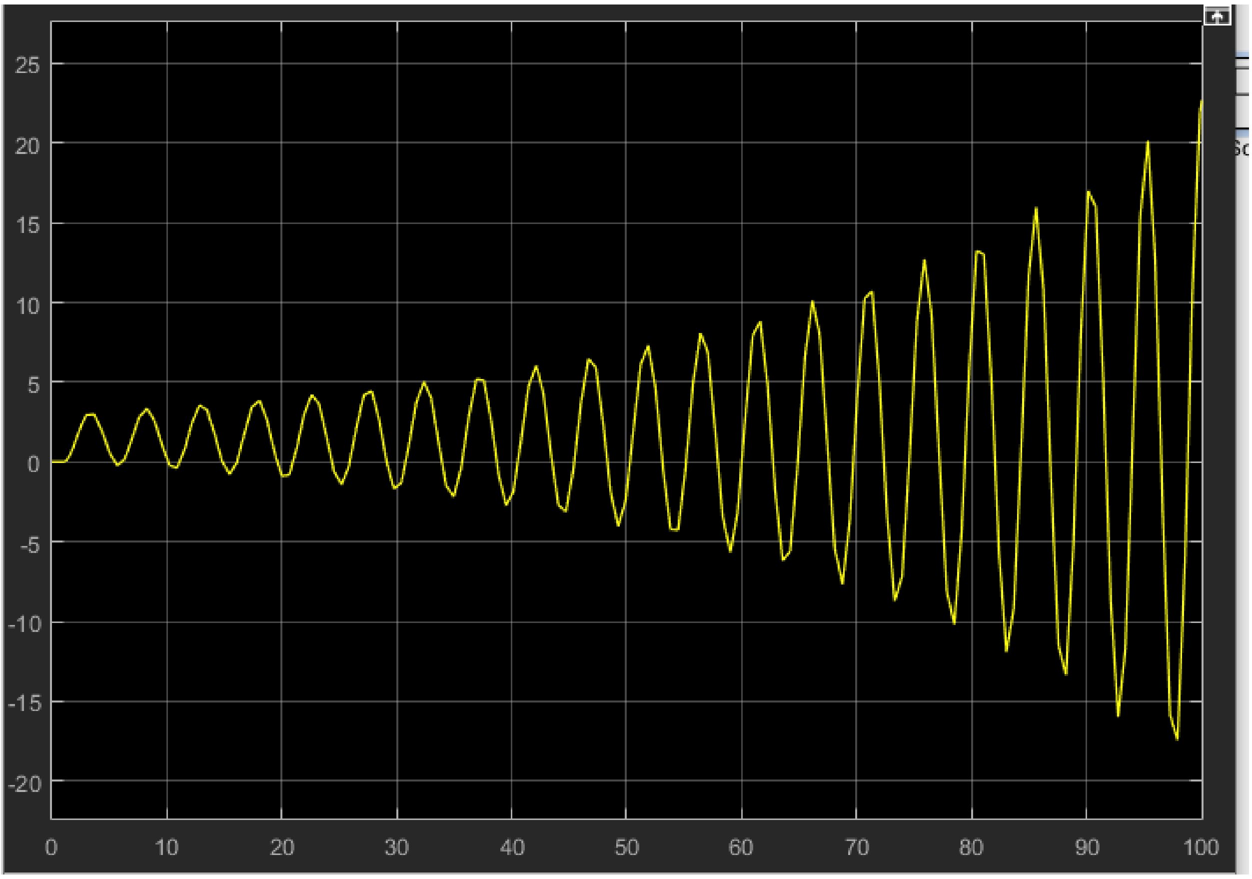
```
h1_tf = tf([2], [1, 1, 6]);
h2_tf = tf([1, 2], [1, 1, 6]);
h3_tf = tf([1, -2], [1, 1, 6]);
h4_tf = tf([1, -5], [1, 1, 6]);
h5_tf = tf([1, 10, 25], [1, 1, 6]);
figure;
hold on;
step(h1_tf);
step(h2_tf);
step(h3_tf);
step(h4_tf);
step(h5_tf);

legend('zero none', 'zero -2', 'zero 2', 'zero 5', 'zero -5', 'Location', 'best');
```

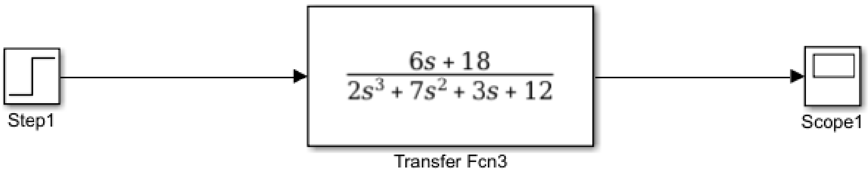


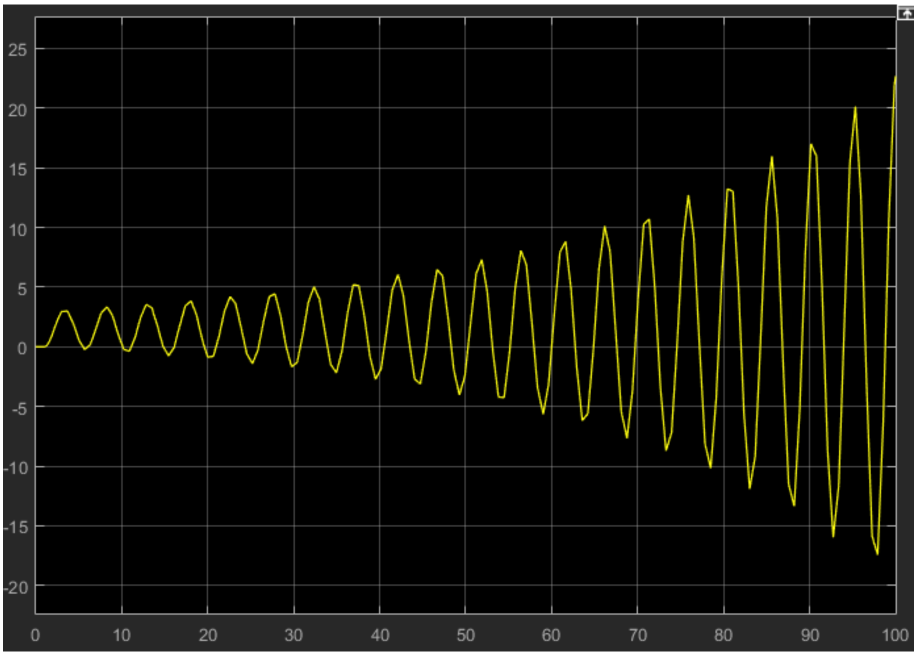
Семинар 2

1.Расчет с помощью MatLAB

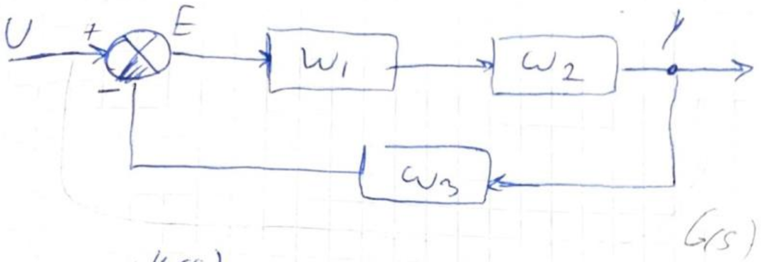








2. Ручной расчет



$W_1(s) = \frac{2}{s}$

$W_2(s) = \frac{3}{2s+1}$

$W_3(s) = \frac{2}{s+3}$

$$W(s) = \frac{Y(s)}{U(s)}$$

$\frac{Y(s)}{U(s)}$ - ?

$E(s) = U(s) - Y(s)$

$Y(s) = E(s)\cdot$

$$W(s) = \frac{W_1 \cdot W_2}{1 + W_1 W_2 \cdot W_3} = \frac{W_1 W_2}{1 + W_1 W_2 W_3}$$

$$W(s) = \frac{\frac{2}{s} \cdot \frac{3}{2s+1}}{1 + \frac{2}{s} \cdot \frac{3}{2s+1} \cdot \frac{2}{s+3}} = \frac{6s+18}{2s^3+7s^2+3s+12}$$

$$U \longrightarrow \boxed{W(s)} \longrightarrow Y$$

$$\frac{Y(s)}{U(s)} = W(s)$$

$$\frac{Y(s)}{U(s)} = \frac{6s+18}{2s^3+7s^2+3s+12}$$

Семинар 3

Расчеты

$$\mathcal{L}\left\{\begin{array}{l}\ddot{y} + \dot{y} + 6y = 2r(t) \\ (s^2 + s + 6)Y(s) = 2R(s)\end{array}\right.$$

$$H(s) = \frac{2}{s^2 + s + 6}$$

$$G(s) = \frac{\overbrace{\frac{k_D s^2 + k_p s + k_I}{s} \cdot \frac{2}{s^2+s+6}}^{1}}{\underbrace{1 + \frac{k_D s^2 + k_p s + k_I}{s} \cdot \frac{2}{s^2+s+6}}_{2}} =$$

$$1\ \frac{2k_D s^2 + 2k_p s + 2k_I}{s^3 + s^2 + 6s}$$

② $$\frac{s^3+s^2+6s+2k_D s^2+2k_p s+2k_I}{s^3+s^2+6s}$$

$$G(s) = \frac{2k_D s^2+2k_p s+2k_I}{\cancel{s^3+s^2+6s}} \cdot \frac{\cancel{s^3+s^2+6s}}{s^3+s^2+6s+2k_D s^2+2k_p s+2k_I} =$$

$$= \frac{2k_D s^2+2k_p s+2k_I}{s(s^2+s+6)+2k_D s^2+2k_p s+2k_I} =$$

$$= \frac{2k_D s^2+2k_p s+2k_I}{\underbrace{s^3+(1+2k_D)s^2+(6+2k_p)s+2k_I}_{=(s+2)^3}}$$

$$s^3+6s^2+12s+8$$

$$\left.\begin{aligned} 1+2k_D &= 6 \\ 6+2k_p &= 12 \\ 2k_I &= 8 \end{aligned}\right| \Rightarrow \begin{aligned} k_D &= 2{,}5 \\ k_p &= 3 \\ k_I &= 4 \end{aligned}$$

$$s^3+(1+2\cdot 2{,}5)s^2+(6+2\cdot 3)s+2\cdot 4 =$$

$$= s^3+6s^2+12s+8$$

$$5\cdot 2{,}5\cdot s^2+5\cdot 3\cdot s+5\cdot 4 = 12{,}5s^2+15s+$$

$$\frac{12{,}5s^2+15s+20}{s^3+6s^2+12s+8}$$

II $W'(s) = \dfrac{12{,}5s^2+15s+20}{s^3+6s^2+12s+8} \cdot \dfrac{1}{12{,}5s^2+15s+20} =$

$= \dfrac{12{,}5s^2+15s+20}{12{,}5x^5+90x^4+260x^3+400x^2+360x+160}$

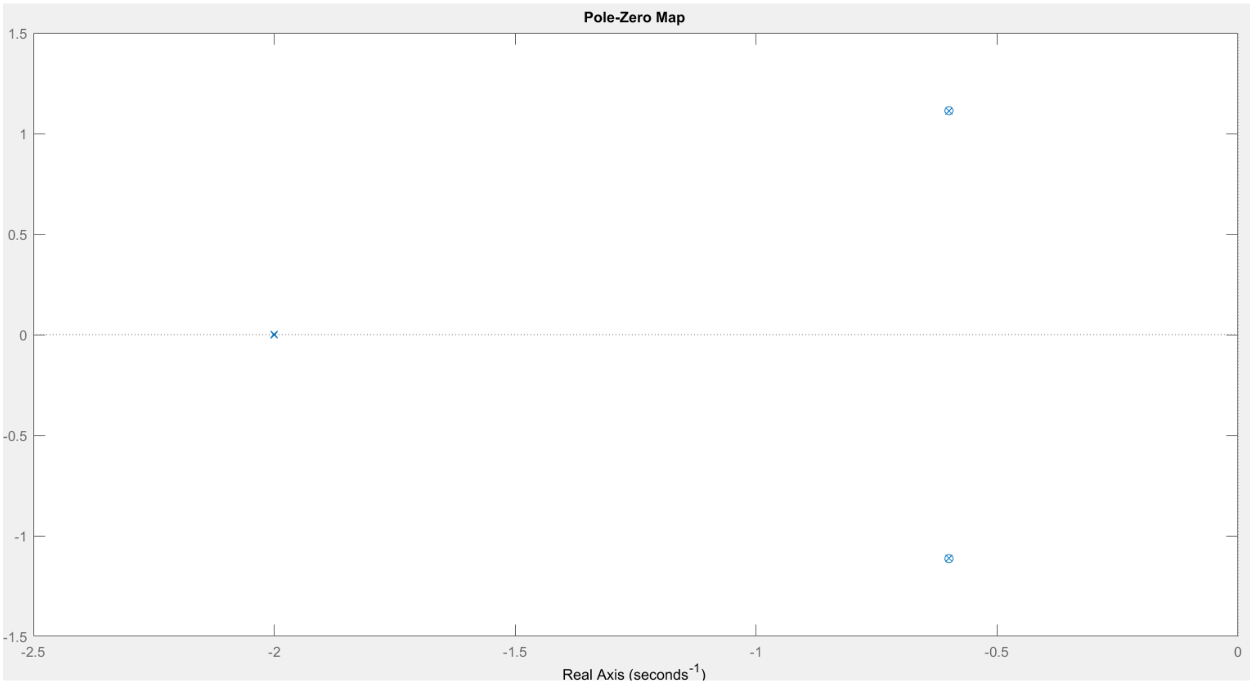
$= \dfrac{\quad}{s^3+6s^2+12s+8}$

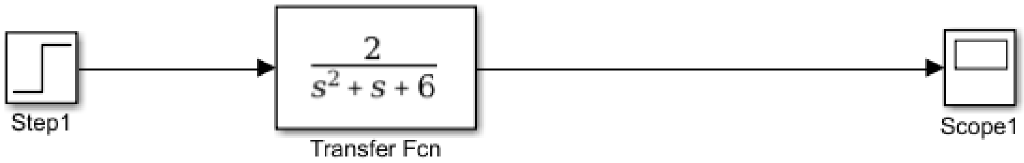
III $y_\infty = \lim\limits_{t\to\infty} y(t) = \lim\limits_{s\to 0} Y(s)\cdot s = \lim\limits_{s\to 0} s \cdot R(s)\cdot W(s) =$

$= \lim\limits_{s\to 0} s\cdot\dfrac{1}{s}\cdot W(s) = \lim\limits_{s\to 0} W(s) = \dfrac{1}{8}$

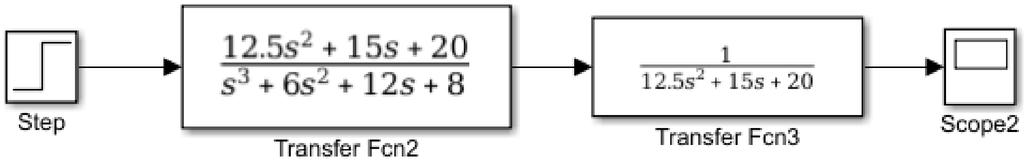
IV $K_r = \dfrac{r(\infty)}{y_\infty} = \dfrac{1}{\frac{1}{8}} = 8$

Картка полюсов

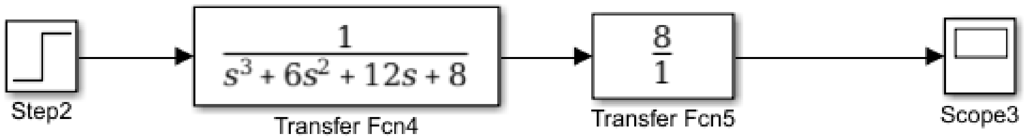




Step1
$\frac{2}{s^2+s+6}$
Transfer Fcn
Scope1

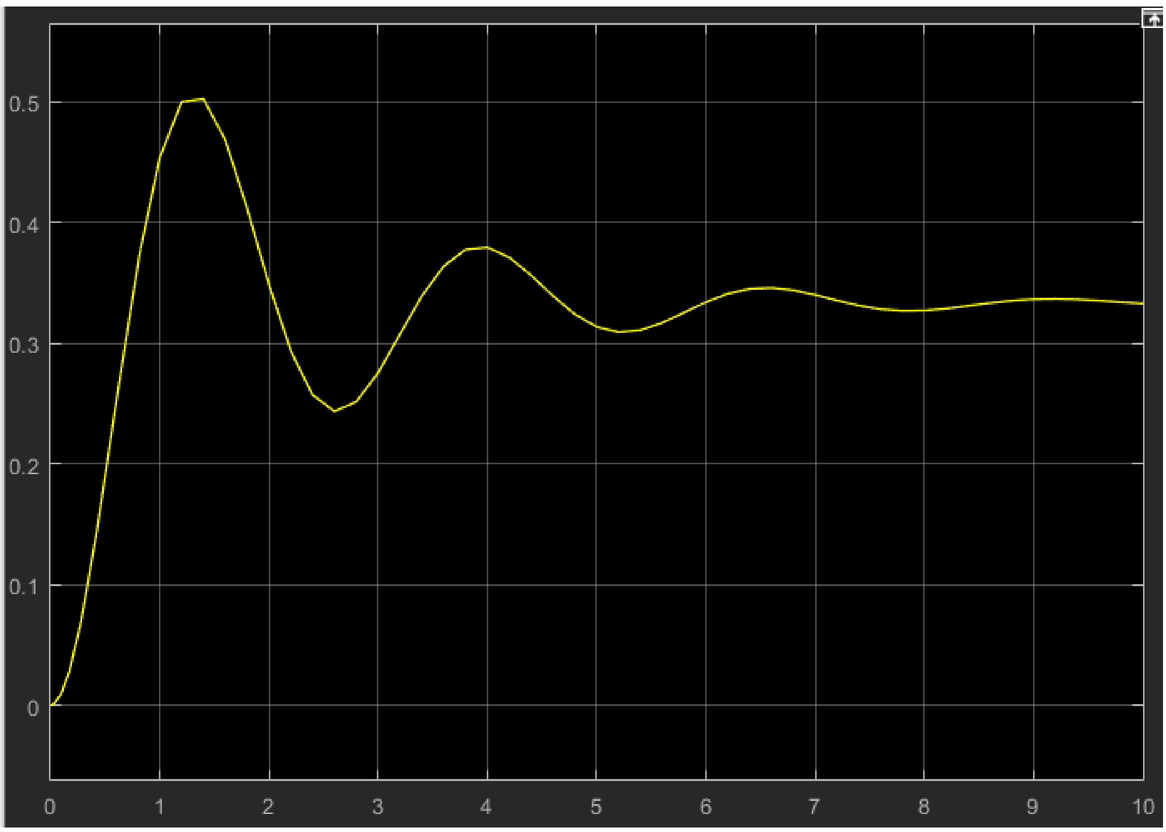


Step
$\frac{12.5s^2+15s+20}{s^3+6s^2+12s+8}$
Transfer Fcn2
$\frac{1}{12.5s^2+15s+20}$
Transfer Fcn3
Scope2



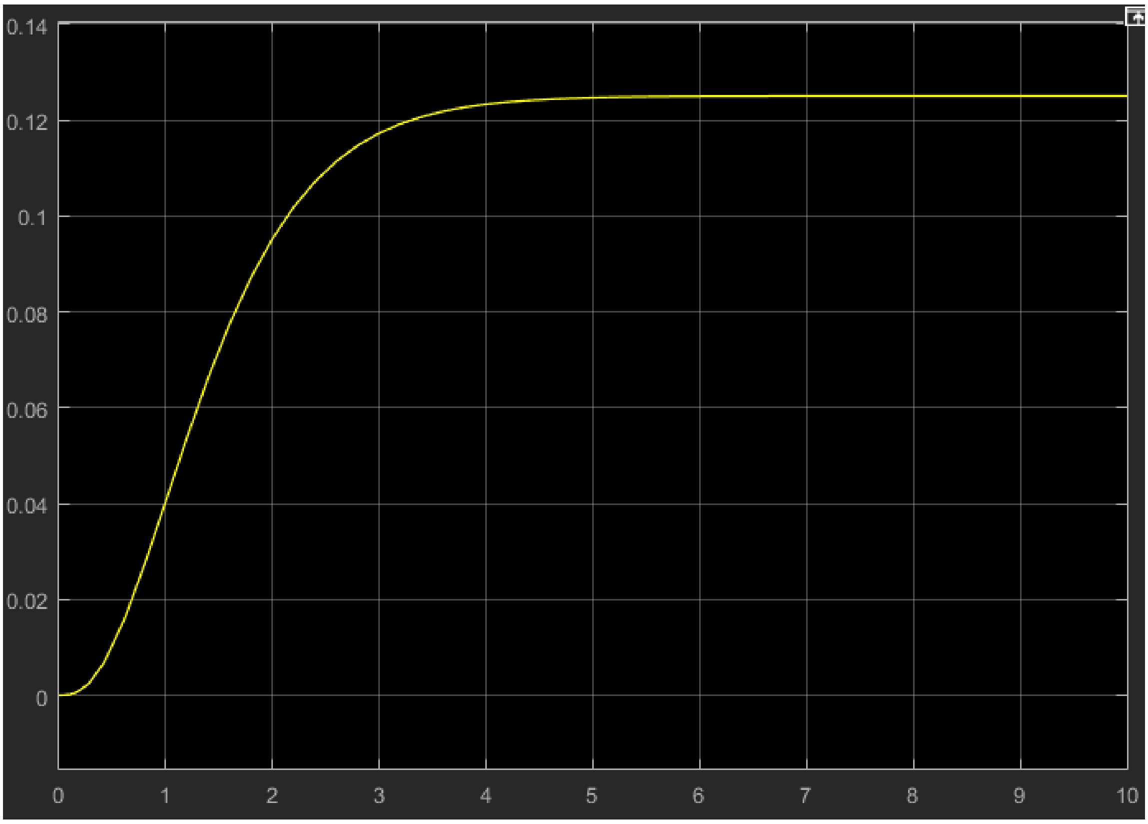
Step2
$\frac{1}{s^3+6s^2+12s+8}$
Transfer Fcn4
$\frac{8}{1}$
Transfer Fcn5
Scope3

Sim1

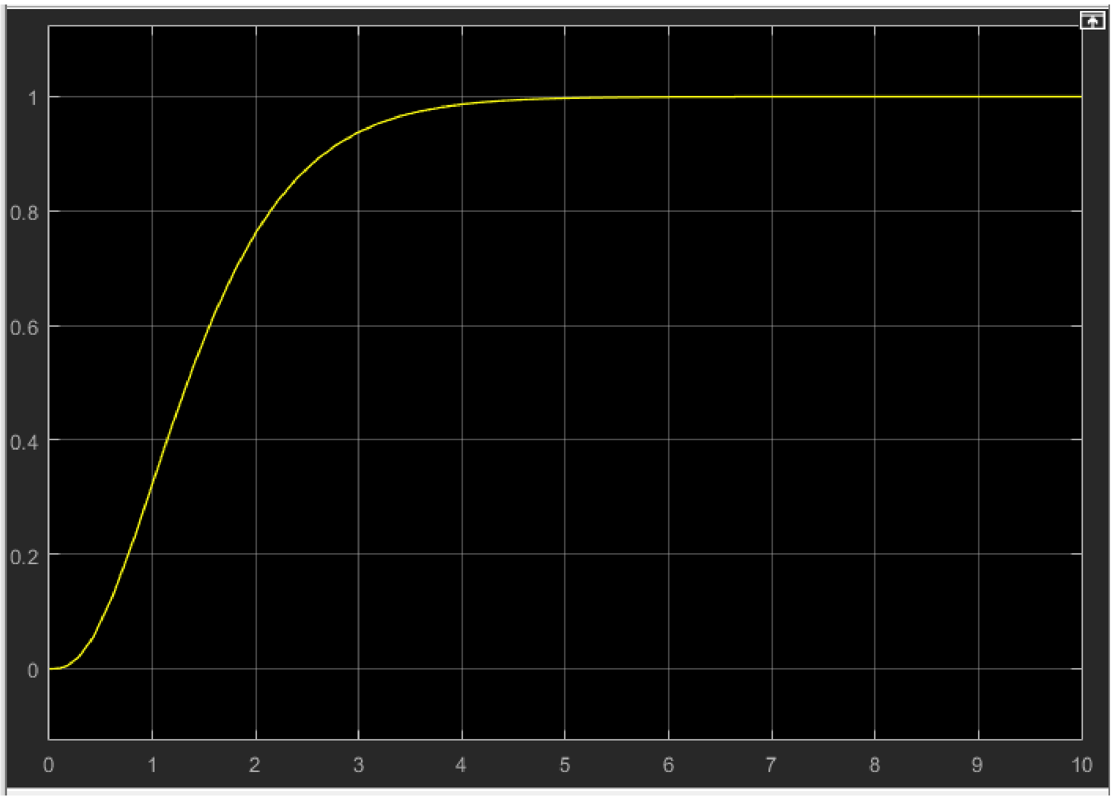


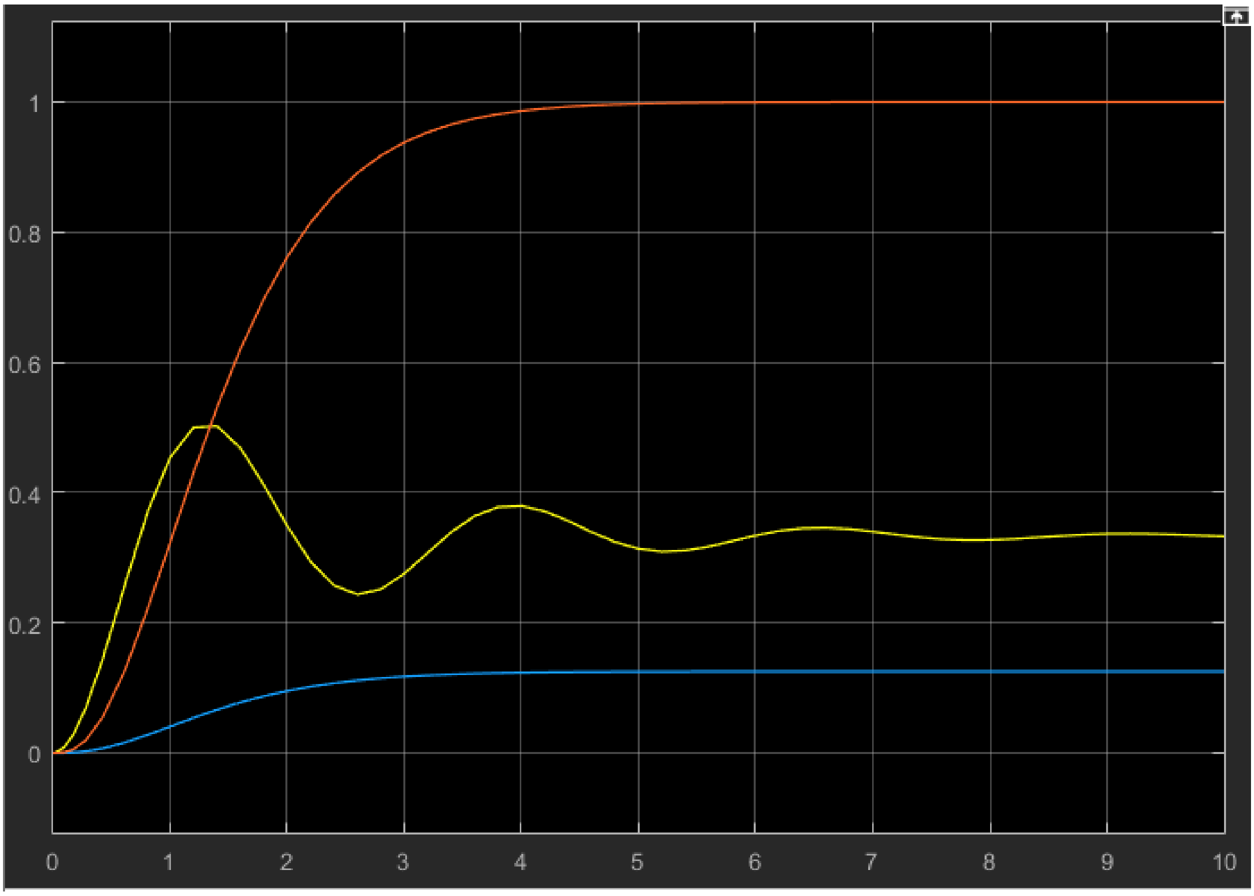
0.5
0.4
0.3
0.2
0.1
0
0
1
2
3
4
5
6
7
8
9
10

Sim2


0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0 1 2 3 4 5 6 7 8 9 10

Sim3


1
0.8
0.6
0.4
0.2
0
0 1 2 3 4 5 6 7 8 9 10


1
0.8
0.6
0.4
0.2
0
0
1
2
3
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9
10